

Chirayu Agrawal

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Education

Shiv Nadar University
Bachelor of Technology in Computer Science and Engineering, GPA: 8.09

Manav Rachna International School
12th Grade: 90%, 10th Grade: 96.4% | Editorial Board Member

Aug. 2022 – May 2026
Noida, UP, India

April 2019 – April 2022
Gurugram, Haryana, India

Professional Experience

Analyst Intern, Enterprise Technology CoE
Bain & Company
Jan. 2026 – Present
Gurugram, Haryana, India

- Architected an AI-enabled full-stack technology intelligence platform using **Python, FastAPI, React**, and **TypeScript** to analyze company technographics; integrated **LLM**-based extraction via **OpenAI API**, semantic clustering, structured output validation, and automated classification to surface actionable insights through a **Next.js** dashboard with real-time **REST API** monitoring.
- Built a scalable analytics pipeline using **Python, DuckDB, SQL**, and **Parquet** to process high-volume operational datasets (1M+ rows), automating multi-sheet **Excel** reporting and reducing end-to-end analysis time by up to **90%**.
- Deployed a suite of enterprise **GenAI** agents using **Python** and the **OpenAI SDK** with multi-threaded batch processing to automate analysis across confidential enterprise workflows, reducing manual effort by **90%** and improving throughput across **100K+** records.
- Technologies used:** Python, FastAPI, React, TypeScript, Next.js, OpenAI API, DuckDB, SQL, Parquet, Excel Automation

Software Engineer Intern
Harman International
May 2025 – Aug. 2025
Gurugram, Haryana, India

- Built an autonomous **AI agent** for automated bug fixing using **CrewAI**; retrieved **Jira** issues, performed root cause analysis, and suggested code-level fixes via semantic search over **GitHub** repositories, achieving **60%** reduction in manual debugging time with **95%** accuracy across 100+ code files.
- Constructed **A2A** servers using Google's **A2A protocol** and integrated multiple agents into a master controller to streamline multi-agent workflows.
- Designed a **YOLO v10** computer vision model for black-screen detection in automotive stereo systems, achieving **92%** accuracy on a 1,000+ image dataset with real-time inference.
- Technologies used:** Python, CrewAI, A2A Protocol, Azure GPT-4o, FAISS, YOLO v10, Jira/GitHub APIs, Sentence Transformers

AI and ML Intern
AgBe Technologies
June 2024 – Aug. 2024
Gurugram, Haryana, India

- Built and deployed a **LightGBM**-based lead scoring model on **50K+** historical leads with engineered demographic, behavioral, and **CRM** features, improving qualified lead conversion rates by **23%**.
- Improved model prediction accuracy by **15%** through advanced feature engineering (interaction features, recency metrics) and data cleaning (outlier removal, missing value imputation).
- Technologies used:** Python, SQL, LightGBM, Scikit-learn, Pandas

Projects & Contributions

SCODA: Self-supervised Continual Domain Adaptation | *Python, PyTorch, Machine Learning*

- ICVGIP 2025 Published* – First-author research proposing a novel **Source-Free Domain Adaptation** framework combining **geometric manifold alignment** with a dual-speed **EMA** teacher-student architecture.
- Achieved **+11.79%** improvement over state-of-the-art on the **OfficeHome** benchmark while preventing catastrophic forgetting.

Open Source Contributor – Microsoft Azure PyRIT | *Python, PyTorch, GenAI*

- Enhanced **AI safety** evaluations by integrating the complete **JailbreakBench (JBB)** dataset into Microsoft's **PyRIT** framework, improving resilience testing against adversarial jailbreak attacks.
- Created a robust data pipeline with harm-category filtering, rich **metadata embedding**, and comprehensive error handling for seamless dataset integration.

Fraud Detection System | *AWS (S3, Kinesis, Lambda, QuickSight)*

- Implemented a real-time, serverless data pipeline using **AWS S3, Kinesis**, and **Lambda** to process transactional data for fraud analysis.
- Produced interactive dashboards in **Amazon QuickSight** to visualize anomalies and fraud trends, improving detection efficiency and stakeholder visibility.

CereBRO | *React, Node.js, LangChain, FastAPI*

- Built an **AI-driven** recommender system and adaptive learning platform for neurodiverse students, delivering personalized learning paths and real-time feedback.
- Leveraged **LangChain** and **Groq API** to generate interactive flashcards and quizzes, increasing student engagement by **25%**.

Technical Skills

Machine Learning: PyTorch, TensorFlow, Scikit-learn, LightGBM, YOLO v10, Transformers, ViT, U-Net, CNNs, Sentence Transformers

LLMs & GenAI: LangChain, OpenAI SDK, RAG, FAISS, Groq API, CrewAI, Multi-agent Systems, OCR

Backend & Data: Python, FastAPI, DuckDB, SQL, Parquet, Pandas, NumPy, REST APIs, Excel Automation

Frontend & Cloud: React, TypeScript, Next.js, AWS (S3, Lambda, Kinesis, SageMaker), Docker, Git, Linux

Languages: Python, Java, JavaScript, C